POWER BI · SQL SERVER · DATA ANALYTICS

OLIST E-COMMERCE ANALYTICS

From raw orders to decision-ready business insights

An end-to-end BI portfolio case study

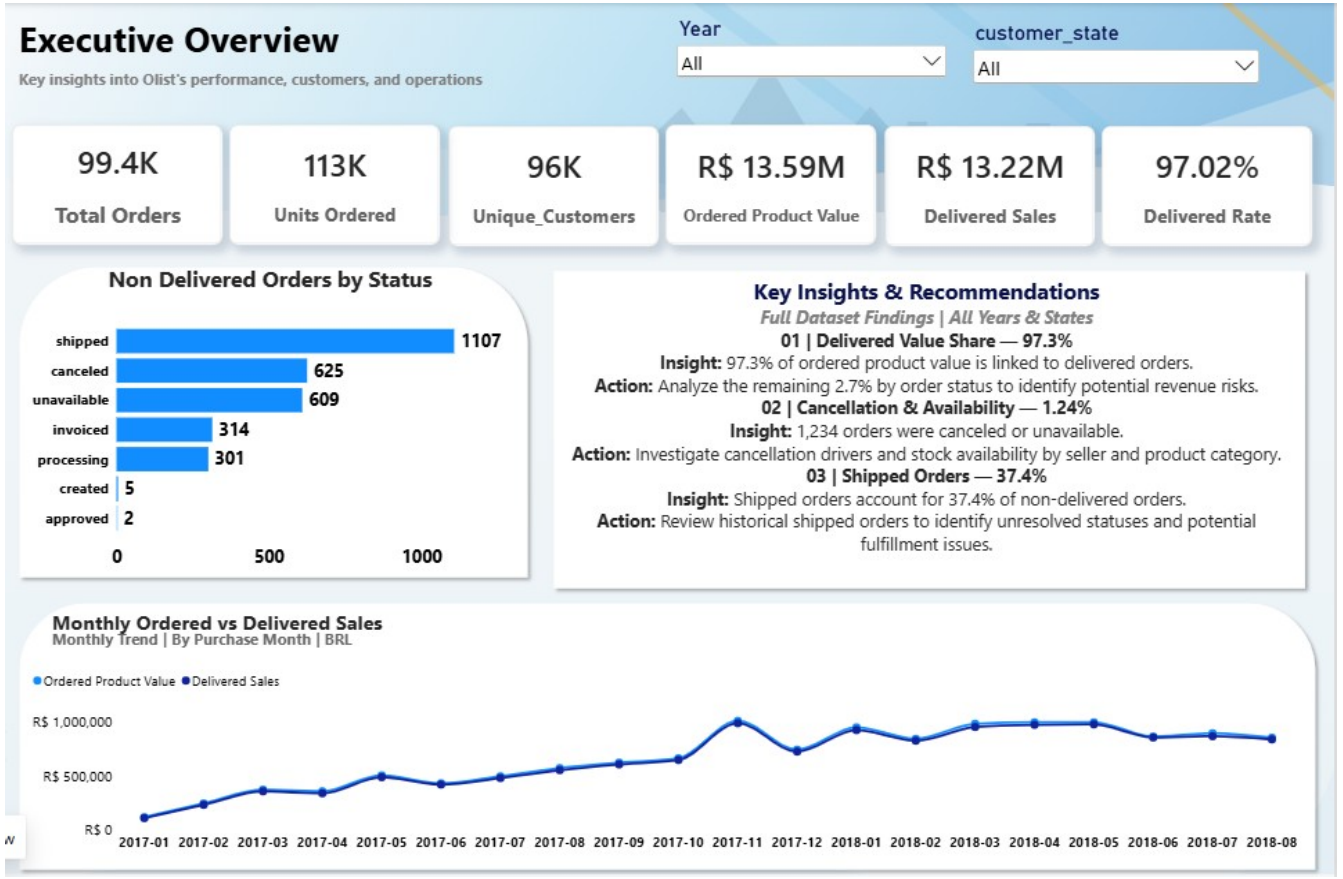
PRESENTED BY MENNA AHMED AMER

GitHub repository: Ecommerce-Sales-Operations-Analytics

9 SOURCE DATASETS

6 SQL VIEWS

6 ANALYTICAL PAGES



Dashboard overview · Six connected analytical views

Executive summary

What the analysis tells us at a glance

99,441
Total orders
Full order-status universe

97.02%
Delivered rate
96,478 delivered orders

R$13.22M
Delivered product sales
Excludes freight charges

3.12%
Repeat-customer rate
All-status orders in observed history

THREE SIGNALS FOR DECISION-MAKERS

1,107
Unconfirmed shipped orders
Largest non-delivered status; investigate final outcomes.

93.23%
On-time delivery
Late delivery links to substantially worse review scores.

78.34%
Credit-card payment share
Recorded payment value is concentrated in one method.

Important: Ordered product value, delivered product sales, and recorded payment value have different definitions.

Business questions & objectives

The reporting layer was designed around operational decisions—not charts alone

01 ORDERS & SALES

Which products and categories drive delivered sales?

Status · Product mix · Trend

02 CUSTOMERS & GEOGRAPHY

Who buys, where, and how frequently?

Identity · Repeat purchase · States

03 FULFILLMENT & REVIEWS

How does delivery performance relate to ratings?

On-time · Late orders · Satisfaction

04 SELLERS & PAYMENTS

Which sellers and payment patterns carry exposure?

Completion · Concentration · Method mix

PROJECT OUTCOME → A six-page interactive Power BI decision-support report

Nine source datasets · 1,550,922 rows

Distinct grains across order, item, customer, payment, review and geolocation data

Orders
99,441
One row per order

Customers
99,441
Customer ID per order

Order items
112,650
One item within an order

Payments
103,886
One payment record

Reviews
99,224
One review record

Products
32,951
One product

Sellers
3,095
One seller

Geolocation
1,000,163
Multiple rows per ZIP prefix

Category translation
71
One category mapping

GRAIN RULE Order, order-item and payment rows are not interchangeable. Join and aggregate at the correct level.

SQL ETL & data-quality workflow

Quality checks before views, relationships, or visual conclusions

01 EXTRACT

Load 9 raw CSV tables

02 PROFILE

Keys, nulls, data types, outliers

03 TRANSFORM

Business fields in 6 SQL views

04 VALIDATE

Reconcile output grains and row counts

DATA QUALITY CONTROLS

Uniqueness & orphan checks	→	Validate PKs, composite keys, links and missing order items.
Date and status eligibility	→	Flag invalid date sequences and delivery-status exceptions.
Numeric quality and outliers	→	Check zero payments and installments; profile P99 values.
Grain-aware analytical views	→	Retain separate order, item, payment and review-level facts.

Observed exceptions include 8 delivered orders missing delivery dates, 775 orders without item rows and 9 zero-value payments.

Orders → vw_Orders_Analysis

Status-aware delivery metrics, date sequences, and eligible delivery calculations

ROWS	DELIVERED	MISSING DELIVERY
99,441	96,478	8 orders

SOURCE ISSUE / VALIDATION	SQL TRANSFORMATION / HANDLING
Orders not always completed	→ Preserve order_status for the whole status universe.
Delivered but missing delivery timestamp	→ Return NULL for ineligible delivery-duration calculations.
Possible invalid carrier/customer chronology	→ CASE flag when customer-delivered precedes carrier date.
Delivery duration in the raw table	→ Derive Delivery_Days from purchase to customer delivery.
Comparing promised vs actual delivery	→ Classify On-Time / Late / Not Applicable by date eligibility.
Late-day measurement	→ Derive Delay_Days only for valid late delivered orders.

DATA RULE Order completion and on-time eligibility are distinct denominators.

Order items → vw_Order_Items_Analysis

Product sales, seller links, translated categories and item-level fulfillment flags

ITEM ROWS	NO-ITEM ORDERS	TRANSLATIONS
112,650	775	71 rows

SOURCE ISSUE / VALIDATION		SQL TRANSFORMATION / HANDLING
Multiple items per order	→	Keep (order_id, order_item_id) composite item grain.
Status is stored at order level	→	JOIN each item to its parent order_status and purchase date.
Portuguese product-category names	→	LEFT JOIN products and translation lookup; expose Category_English.
Missing category translations	→	COALESCE unknown categories rather than discard the item.
Delivered vs all-item scope	→	Derive Is_Delivered flag from the order-level status.
Potential payment join fan-out	→	Keep payment analysis separate; aggregate before cross-fact joins.

DATA RULE A no-item order cannot contribute product sales, but it remains in order counts.

Customers → vw_Customers_Analysis

Distinguish customer records from unique people, then build repeat behavior

CUSTOMER IDS	UNIQUE PEOPLE	REPEAT RATE
99,441	96,096	3.12%

SOURCE ISSUE / VALIDATION		SQL TRANSFORMATION / HANDLING
Two identifiers with different meaning	→	customer_id is an order-level link; customer_unique_id identifies a person.
Multiple orders by the same person	→	JOIN customers to orders using customer_id.
Need comparable customer totals	→	GROUP BY customer_unique_id to construct customer-level rows.
Repeat purchase logic	→	SQL view flags >1 delivered orders; Power BI reports all-order repeat.
First and last activity per person	→	MIN / MAX order_purchase_timestamp for purchase span.
Risk of overstating loyalty	→	3.12% uses all orders in history; not a retention cohort.

DATA RULE The SQL delivered-only repeat flag differs from the all-order 3.12% Power BI KPI.

Payments → vw_Payments_Analysis

Payment rows, valid installment classifications and monetary definitions

PAYMENT ROWS	ZERO-VALUE	ZERO INSTALLMENTS
103,886	9 records	2 records

SOURCE ISSUE / VALIDATION		SQL TRANSFORMATION / HANDLING
More than one payment per order	→	Retain payment_sequential and order_id as payment grain.
Invalid installment values	→	Set Invalid_Installments_Flag when installments ≤ 0.
Different payment types	→	Classify credit card vs other methods before installment analysis.
One-time vs installment credit-card use	→	Create Credit_Card_Installment_Group from installment count.
Order completion context	→	Join each payment record to order status.
Payment value and product sales differ	→	Do not claim R$16.01M in recorded payments equals net revenue.

QUALITY Keep zero-value records visible for validation; do not treat them as positive sales.

Reviews → vw_Order_Reviews_Analysis

Order-level review aggregation and reliable links to delivery status

REVIEW ROWS	REVIEWED ORDERS	AVG RATING
99,224	98,673	4.09

SOURCE ISSUE / VALIDATION		SQL TRANSFORMATION / HANDLING
One order may have multiple reviews	→	WITH Reviews_Summary aggregates records by order_id.
Repeated review rows inflate joins	→	Create one review summary row per order before modeling.
Need review-coverage context	→	COUNT Review_Count and MIN/MAX review_score.
Average rating per reviewed order	→	AVG CAST(review_score AS FLOAT), then ROUND.
Delivery status & satisfaction	→	Join review summary with vw_Orders_Analysis.
Avoid unsupported cause claims	→	Compare late vs on-time ratings as association only.

QUALITY Rating 4.09 uses all available reviewed statuses; delivery slices use eligible delivered orders.

Geolocation → vw_Geolocation_Clean

Convert raw coordinates, flag invalid bounds and standardize ZIP-prefix geography

RAW GEO ROWS	ZIP PREFIXES	GEO VIEW
1,000,163	19,015	1 row / ZIP

SOURCE ISSUE / VALIDATION		SQL TRANSFORMATION / HANDLING
Latitude / longitude may arrive as text	→	Use TRY_CONVERT(FLOAT) for both coordinate fields.
Geographic coordinates need sanity checks	→	Flag plausible Brazil bounds: lat -35..6; lon -75..-30.
Several coordinate rows per ZIP prefix	→	GROUP BY geolocation_zip_code_prefix.
Avoid invalid-coordinate averages	→	AVG only coordinates marked usable.
Need transparent coverage diagnostics	→	COUNT usable and excluded source records.
Maps require dependable status	→	Create Geo_Status: Usable or No Usable Coordinates.

DATA RULE Geolocation view aggregates repeated ZIP-prefix records rather than creating false customer rows.

Relational model & data flow

Actual Power BI Model View with order, item, customer, payment, review and date paths



GRAIN & FILTER PATHS

ORDER

FactOrders + customer-order link

ITEM

FactOrderItems + product/seller

PAYMENT

FactPayments per payment record

REVIEW

FactReviews aggregated by order

Actual model screenshot

KPI framework & DAX

Measures align technical calculations with decision-maker questions

ORDERS & SALES

- Total Orders
- Delivered Sales (R$)
- Delivered Rate
- Delivered AOV
- YoY / MoM Growth

CUSTOMERS & GEOGRAPHY

- Unique Customers
- Repeat Customers
- Repeat Rate
- Orders per Customer
- Sales per Customer

FULFILLMENT & REVIEWS

- Avg Delivery Days
- On-Time Delivery Rate
- Late Orders
- Avg Days Late
- Avg Review Score

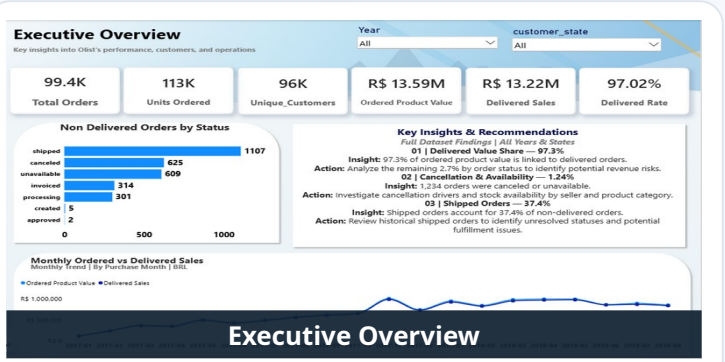
SELLERS & PAYMENTS

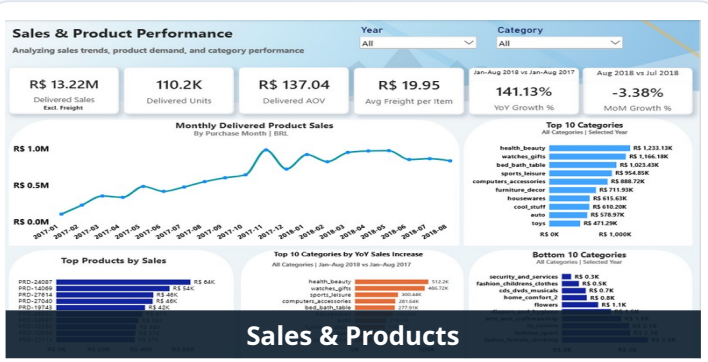
- Seller Completion Rate
- Sellers Below 95% (≥20 orders)
- Total Payment Value
- Multi-Payment Order Rate
- Credit-card Share

KPI design: distinct keys, correct status eligibility, separate monetary scopes and review-aware denominators.

Dashboard architecture & user experience

Six Power BI analytical pages with slicers, consistent KPIs and business questions

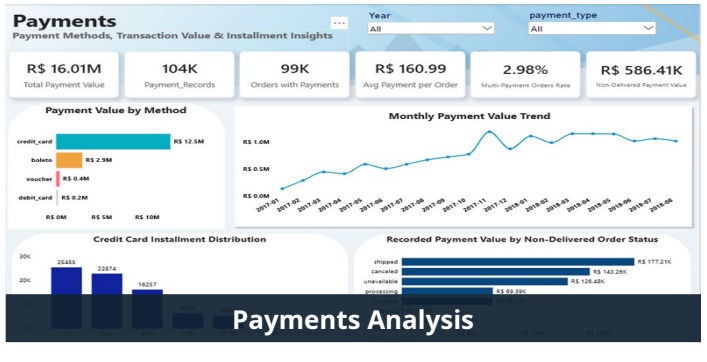






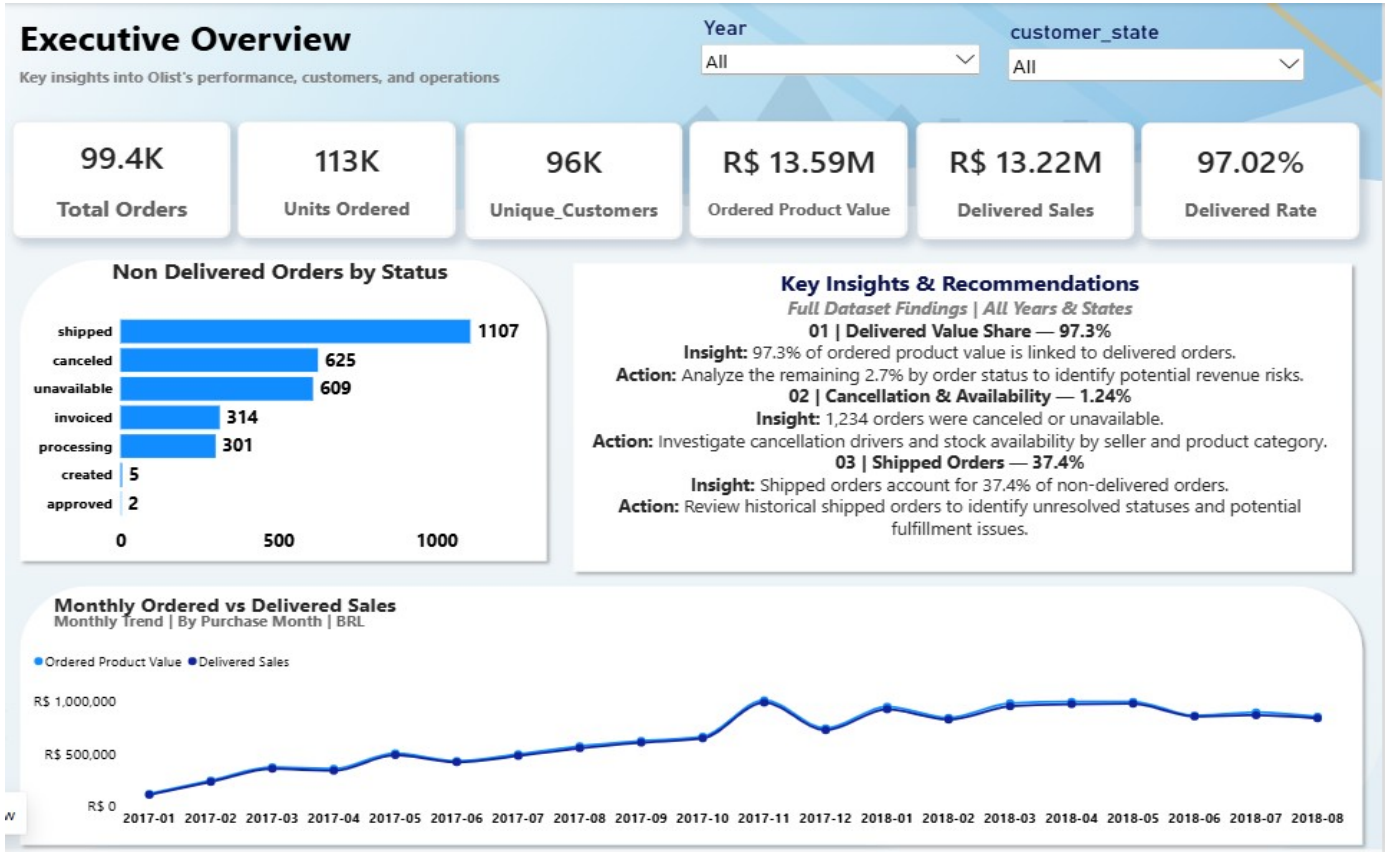






Executive overview

Top cross-functional signals, with actions linked to findings



KEY INSIGHTS & RECOMMENDATIONS

01 INSIGHT

97.02% of 99,441 orders were delivered.

RECOMMENDATION

Keep delivery-status definitions consistent across reports.

02 INSIGHT

1,107 orders remain in shipped status.

RECOMMENDATION

Investigate unconfirmed fulfillment and status updates.

03 INSIGHT

1,234 orders were canceled or unavailable.

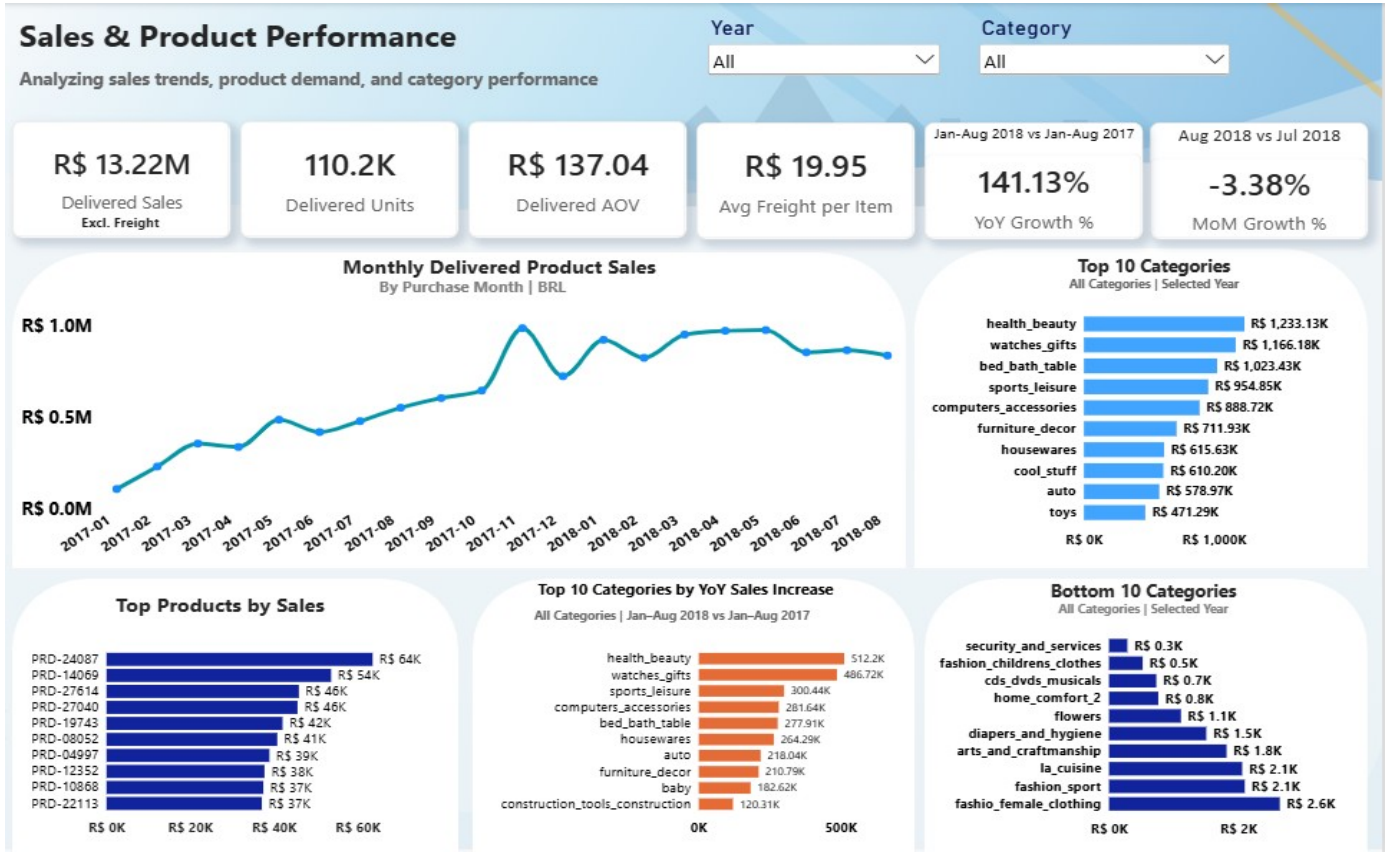
RECOMMENDATION

Examine inventory and cancellation drivers by category.

Delivered sales are product-only; the dashboard is not a P&L or cash-flow statement.

Sales & product performance

Which categories drive sales, and how did sales evolve?



KEY INSIGHTS & RECOMMENDATIONS

01 INSIGHT

Delivered product sales reached R$13.22M.

RECOMMENDATION

Focus decisions on delivered product value, excluding freight.

02 INSIGHT

Jan–Aug 2018 grew 141.13% vs 2017, from a smaller base.

RECOMMENDATION

Check whether the post-March 2018 plateau persists.

03 INSIGHT

Top five categories generate roughly 39.8% of sales.

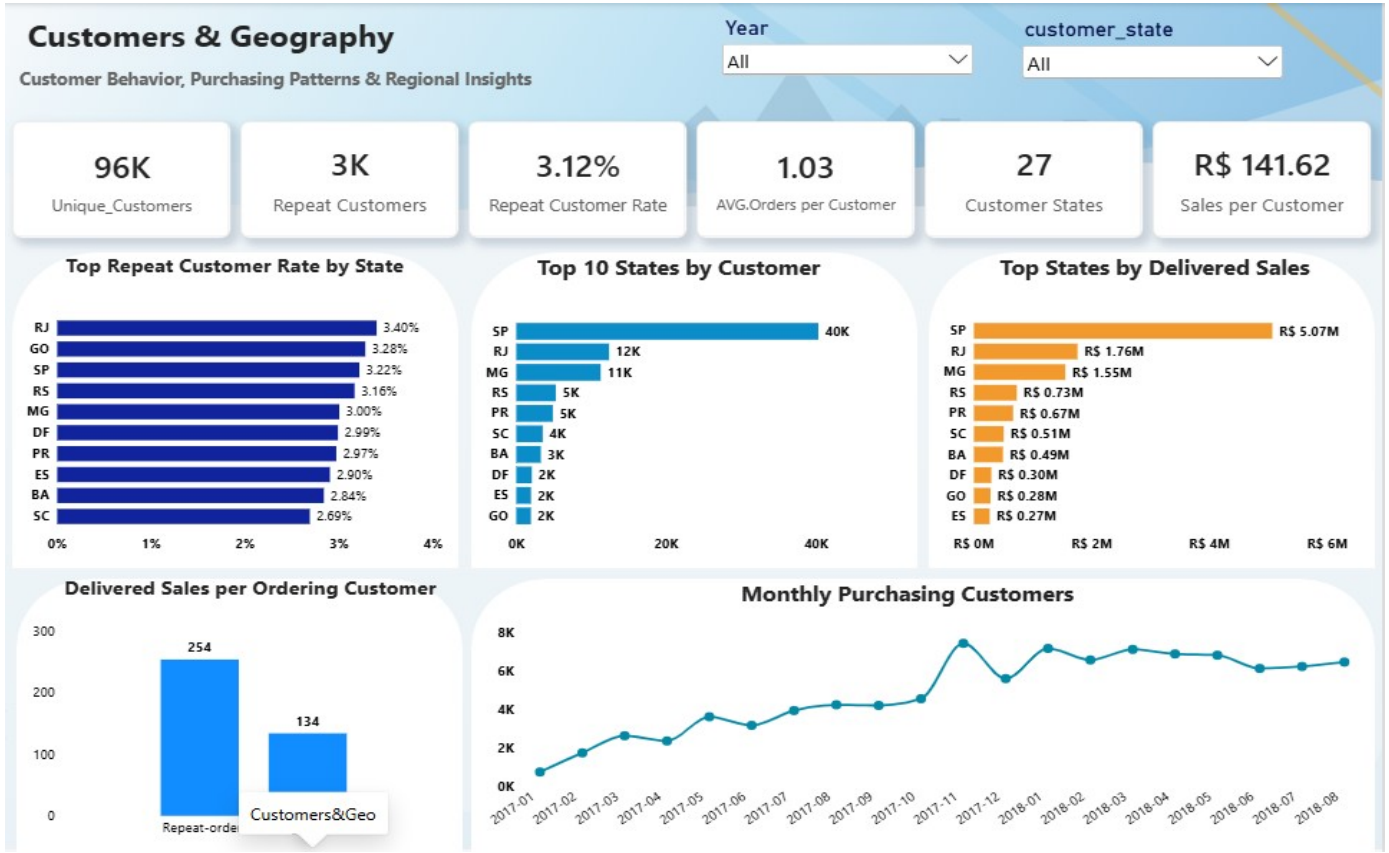
RECOMMENDATION

Monitor concentration and growth in adjacent categories.

YoY uses comparable Jan–Aug windows; growth does not establish a sustainable long-term trend.

Customers & geography

Customer identity, geographic concentration and observed repeat behavior



KEY INSIGHTS & RECOMMENDATIONS

01 INSIGHT

Only 3.12% are repeat customers in observed history.

RECOMMENDATION

Pilot re-engagement by first-purchase category and cohort.

02 INSIGHT

São Paulo represents ~41.9% of customers.

RECOMMENDATION

Monitor demand concentration and service outside SP.

03 INSIGHT

Repeat buyers show higher sales per customer.

RECOMMENDATION

Test whether a repeat-buyer program can lift value.

customer_unique_id identifies people; repeat rate is NOT a cohort retention metric.

Delivery & customer reviews

Operational reliability and the association between delays and satisfaction



KEY INSIGHTS & RECOMMENDATIONS

01 INSIGHT

93.23% of eligible delivered orders arrived on time.

RECOMMENDATION

Monitor late-delivery cohorts and seller/carrier patterns.

02 INSIGHT

On-time orders average ~4.3 stars; late orders ~2.3.

RECOMMENDATION

Prioritize proactive contact for delayed customers.

03 INSIGHT

6,534 orders arrived late (eligible population).

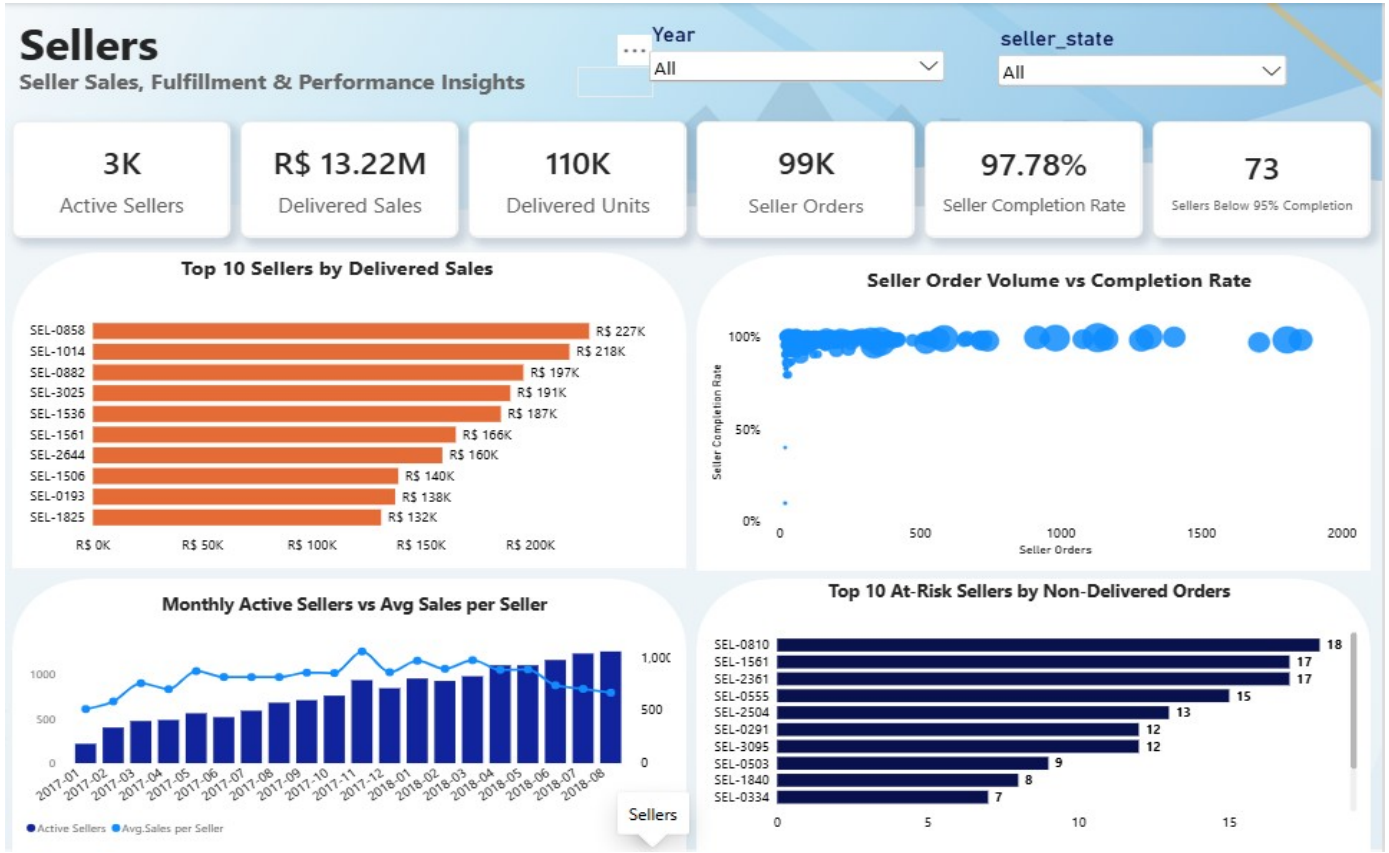
RECOMMENDATION

Investigate high-delay months and high-risk categories.

Rating 4.09 includes all reviewed statuses; timing comparisons use eligible delivered orders.

Seller performance

Completion reliability and exposure among higher-volume sellers



One multi-seller order may count under multiple sellers; seller-order totals are not additive.

KEY INSIGHTS & RECOMMENDATIONS

01 INSIGHT

Seller-linked completion rate is 97.78%.

RECOMMENDATION

Track completion at seller-order grain, not summed order totals.

02 INSIGHT

73 sellers fall below 95% with ≥ 20 orders.

RECOMMENDATION

Review the flagged sellers and priority failure drivers.

03 INSIGHT

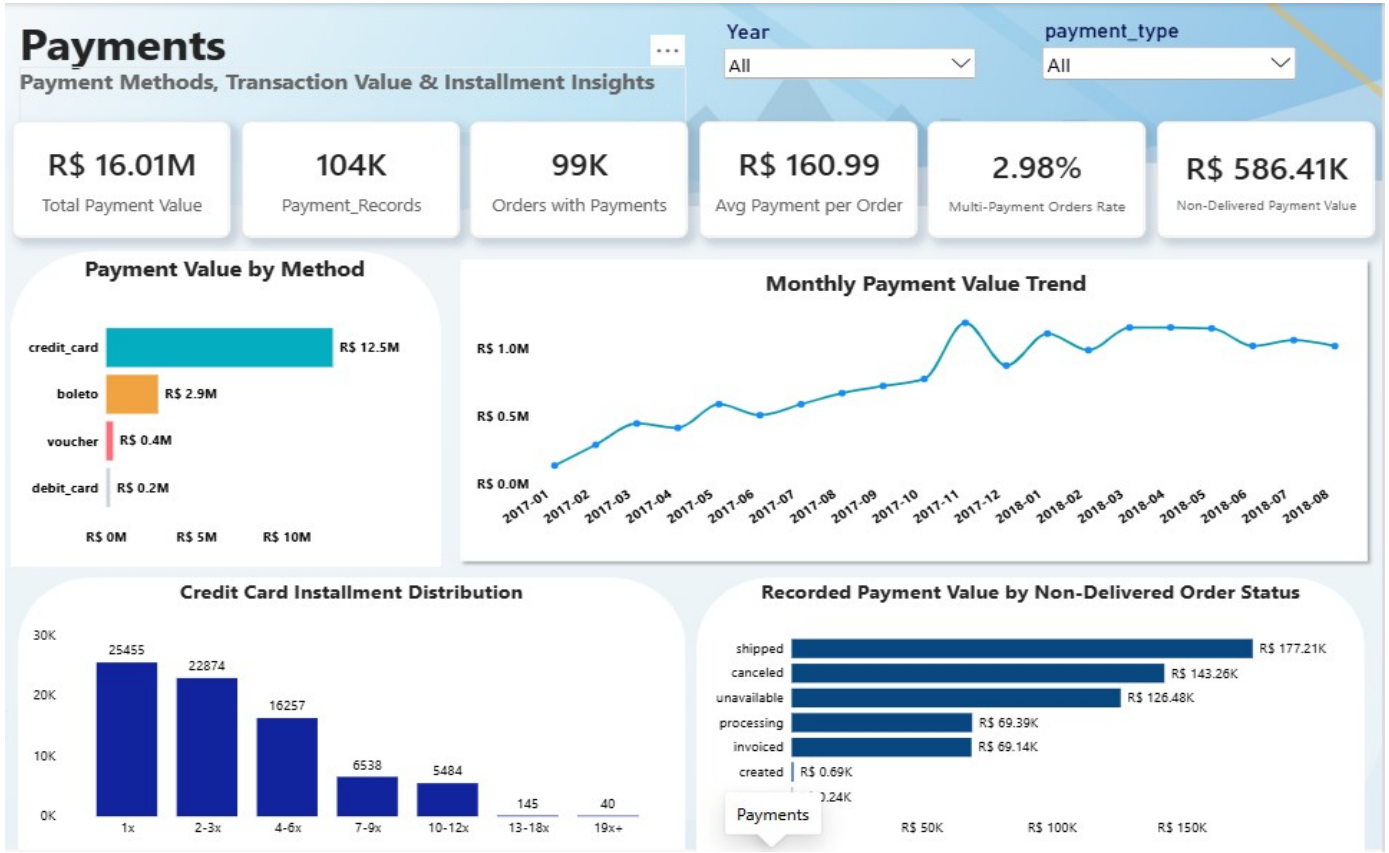
Top ten sellers produce ~13.27% of delivered sales.

RECOMMENDATION

Track concentration alongside merchant service reliability.

Payments & installment analysis

Method mix, multi-payment orders and non-delivered payment exposure



Total payment value R$16.01M is not equivalent to product sales or recognized revenue.

KEY INSIGHTS & RECOMMENDATIONS

01 INSIGHT

Credit cards represent 78.34% of recorded value.

RECOMMENDATION

Track method dependency and reconciliation risks.

02 INSIGHT

2.98% of paid orders have multiple payment rows.

RECOMMENDATION

Aggregate by order before comparing payment and item facts.

03 INSIGHT

R$586.41K is linked to non-delivered orders.

RECOMMENDATION

Audit status and refund settlements before declaring losses.

Four findings that matter

Evidence first. Business implications second. Actions next.

1,107 — **Unconfirmed shipped status**

Largest unresolved non-delivered group.

Validate fulfillment outcomes and update status.

3.12% — **Observed repeat purchases**

Most customers appear only once in window.

Test a retention hypothesis with cohort controls.

4.3 vs 2.3 — **Delivery-linked customer ratings**

Late orders receive much poorer ratings.

Monitor delay risks and proactive support.

78.34% — **Credit-card value concentration**

Recorded payment mix relies on credit cards.

Review payment operations and reconciliation.

Recommendations & measurement plan

Move from observed patterns to testable operational decisions

PRIORITY	EVIDENCE	RECOMMENDED ACTION	KPI TO TRACK
P1	1,107 shipped orders	Reconcile shipment outcomes and status history.	Unresolved shipped count
P1	Late rating ~2.3 vs ~4.3	Test targeted delivery-delay outreach.	Late rate + reviewed rating
P2	3.12% repeat customers	Run retention pilots segmented by first category.	Repeat rate by cohort
P2	73 sellers below 95%	Review seller-level order failures; prioritize high volume.	Seller completion rate
P2	R$586.41K non-delivered payments	Verify settlement and refund outcomes by order.	Unreconciled payments

Recommendations are hypotheses for pilot testing—not confirmed causal effects or already-measured business impact.

From raw data to a decision-ready report

Project scope, important limitations and a practical next step

SOURCE DATA | SQL QUALITY | 6 ANALYTICAL VIEWS | POWER BI MODEL | DAX KPIs | **BUSINESS ACTIONS**

WHAT WAS DELIVERED

9	source datasets	**6**	SQL views
6	dashboard pages	**99.4K**	orders evaluated

Menna Ahmed Amer | Olist Analytics

LIMITATIONS & NEXT STEPS

- Historical public dataset; not Olist company financial reporting.
- Payments, freight and product sales have different scopes.
- Repeat purchase is not full cohort retention or customer LTV.
- Next: test fulfillment and retention pilots with new data.

Repository: Ecommerce-Sales-Operations-Analytics

TURNING E-COMMERCE DATA INTO ACTIONABLE BUSINESS INSIGHTS